



Threagile

Agile Threat Modeling

Threat Model Report

Ship Bridge Navigation

29 May 2026

threatmod-automation

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About Threagile

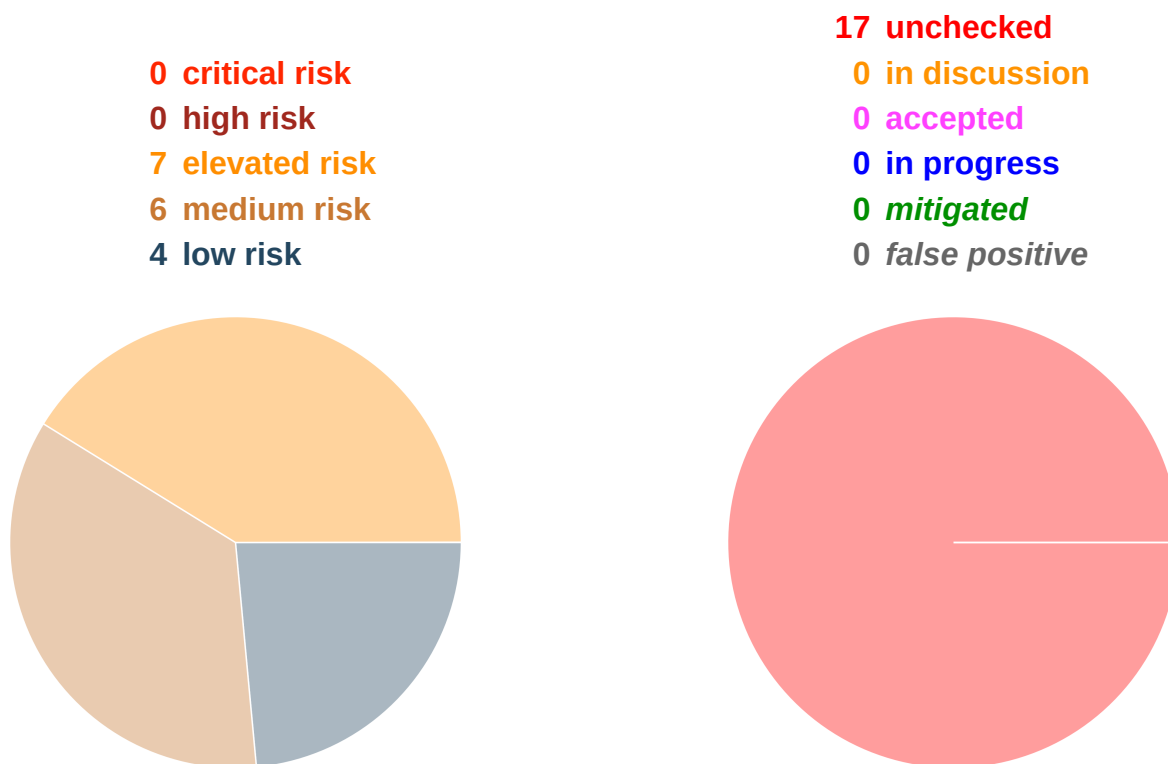
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Management Summary

Threagile toolkit was used to model the architecture of "Ship Bridge Navigation" and derive risks by analyzing the components and data flows. The risks identified during this analysis are shown in the following chapters. Identified risks during threat modeling do not necessarily mean that the vulnerability associated with this risk actually exists: it is more to be seen as a list of potential risks and threats, which should be individually reviewed and reduced by removing false positives. For the remaining risks it should be checked in the design and implementation of "Ship Bridge Navigation" whether the mitigation advices have been applied or not.

Each risk finding references a chapter of the OWASP ASVS (Application Security Verification Standard) audit checklist. The OWASP ASVS checklist should be considered as an inspiration by architects and developers to further harden the application in a Defense-in-Depth approach. Additionally, for each risk finding a link towards a matching OWASP Cheat Sheet or similar with technical details about how to implement a mitigation is given.

In total **17 initial risks** in **9 categories** have been identified during the threat modeling process:



Navigation-at-sea architecture focused on bridge sensors, ECDIS, autopilot control, voyage recording, and the controlled conduit to shore monitoring.

Impact Analysis of 17 Initial Risks in 9 Categories

The most prevalent impacts of the **17 initial risks** (distributed over **9 risk categories**) are (taking the severity ratings into account and using the highest for each category):

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated: [Cross-Site Scripting \(XSS\)](#): 4 Initial Risks - Exploitation likelihood is *Likely with Medium impact*.

If this risk remains unmitigated, attackers might be able to access individual victim sessions and steal or modify user data.

Elevated: [SQL/NoSQL-Injection](#): 3 Initial Risks - Exploitation likelihood is *Very Likely with Medium impact*.

If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Medium: [Cross-Site Request Forgery \(CSRF\)](#): 1 Initial Risk - Exploitation likelihood is *Very Likely with Low impact*.

If this risk remains unmitigated, attackers might be able to trick logged-in victim users into unwanted actions within the web application by visiting an attacker controlled web site.

Medium: [Missing Cloud Hardening](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Medium impact*.

If this risk is unmitigated, attackers might access cloud components in an unintended way.

Medium: [Missing Hardening](#): 4 Initial Risks - Exploitation likelihood is *Likely with Low impact*.

If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Low: [Missing Build Infrastructure](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model due to critical build infrastructure components missing in the model.

Low: [Missing Identity Store](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model in the identity provider/store that is currently missing in the model.

Low: [Missing Network Segmentation](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Low impact*.

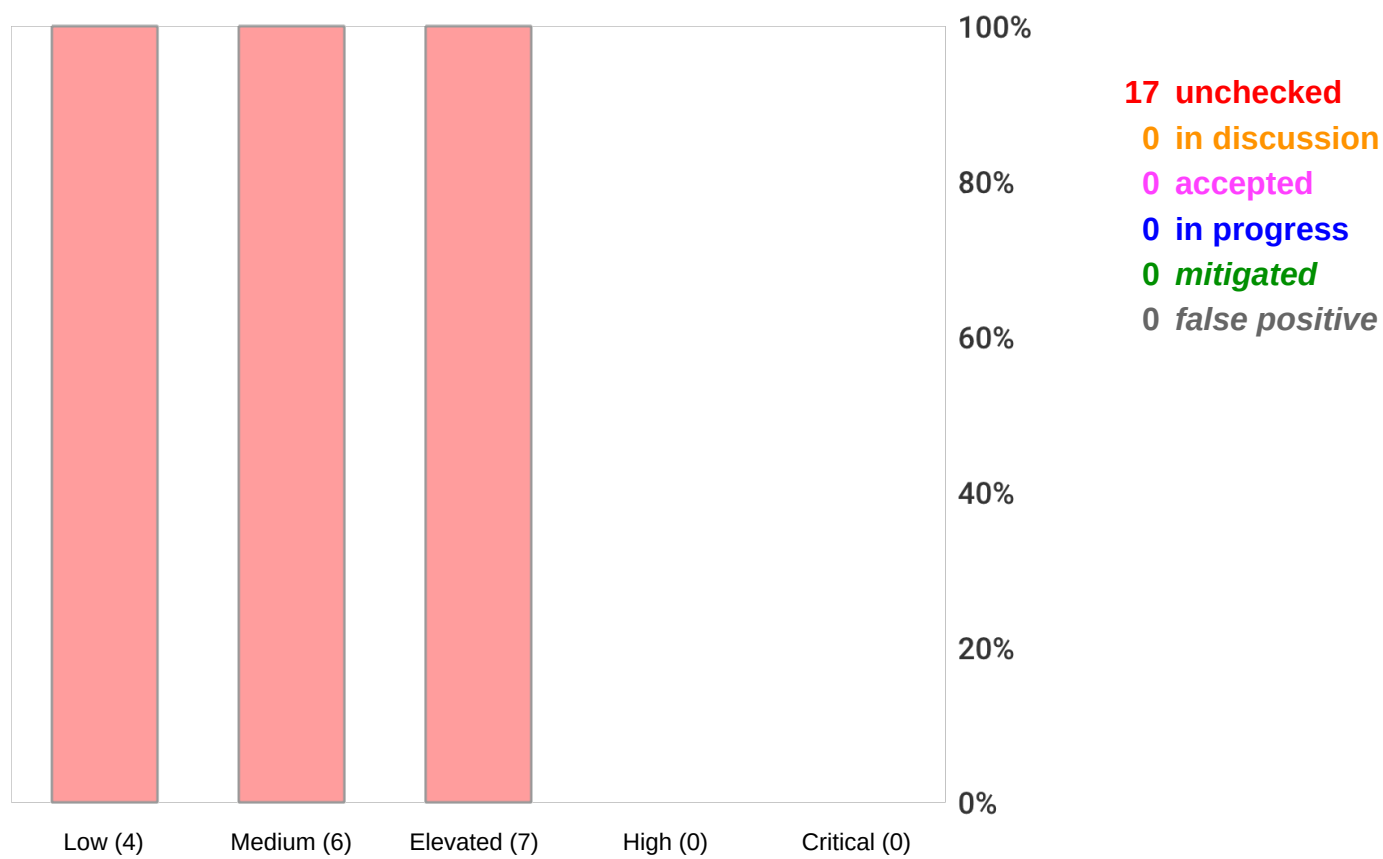
If this risk is unmitigated, attackers successfully attacking other components of the system might have an easy path towards more valuable targets, as they are not separated by network segmentation.

Low: [Missing Vault \(Secret Storage\)](#): 1 Initial Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to easier steal config secrets (like credentials, private keys, client certificates, etc.) once a vulnerability to access files is present and exploited.

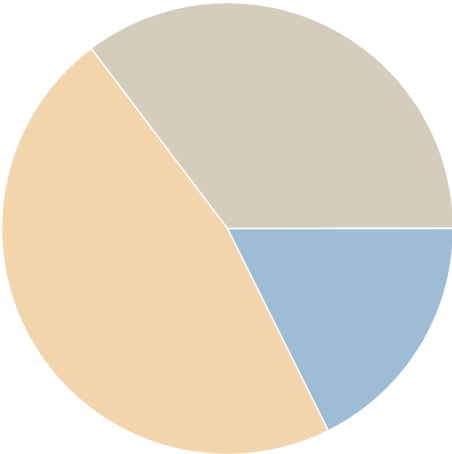
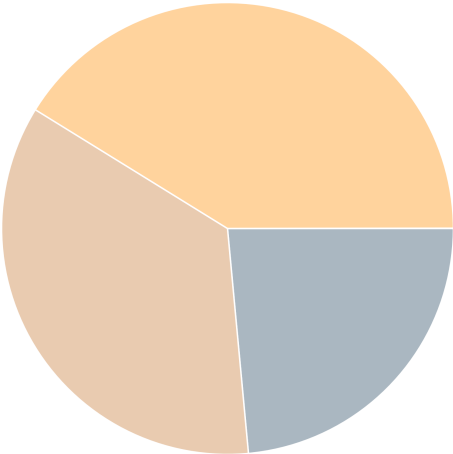
Risk Mitigation

The following chart gives a high-level overview of the risk tracking status (including mitigated risks):



After removal of risks with status *mitigated* and *false positive* the following 17 remain unmitigated:

- 0 unmitigated critical risk
 - 0 unmitigated high risk
 - 7 unmitigated elevated risk
 - 6 unmitigated medium risk
 - 4 unmitigated low risk
- 0 business side related
 - 3 architecture related
 - 8 development related
 - 6 operations related



Impact Analysis of 17 Remaining Risks in 9 Categories

The most prevalent impacts of the **17 remaining risks** (distributed over **9 risk categories**) are (taking the severity ratings into account and using the highest for each category):

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated: [Cross-Site Scripting \(XSS\)](#): 4 Remaining Risks - Exploitation likelihood is *Likely with Medium impact*.

If this risk remains unmitigated, attackers might be able to access individual victim sessions and steal or modify user data.

Elevated: [SQL/NoSQL-Injection](#): 3 Remaining Risks - Exploitation likelihood is *Very Likely with Medium impact*.

If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Medium: [Cross-Site Request Forgery \(CSRF\)](#): 1 Remaining Risk - Exploitation likelihood is *Very Likely with Low impact*.

If this risk remains unmitigated, attackers might be able to trick logged-in victim users into unwanted actions within the web application by visiting an attacker controlled web site.

Medium: [Missing Cloud Hardening](#): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Medium impact*.

If this risk is unmitigated, attackers might access cloud components in an unintended way.

Medium: [Missing Hardening](#): 4 Remaining Risks - Exploitation likelihood is *Likely with Low impact*.

If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Low: [Missing Build Infrastructure](#): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model due to critical build infrastructure components missing in the model.

Low: [Missing Identity Store](#): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model in the identity provider/store that is currently missing in the model.

Low: [Missing Network Segmentation](#): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers successfully attacking other components of the system might have an easy path towards more valuable targets, as they are not separated by network segmentation.

Low: [Missing Vault \(Secret Storage\)](#): 1 Remaining Risk - Exploitation likelihood is *Unlikely with Low impact*.

If this risk is unmitigated, attackers might be able to easier steal config secrets (like credentials, private keys, client certificates, etc.) once a vulnerability to access files is present and exploited.

Application Overview

Business Criticality

The overall business criticality of "Ship Bridge Navigation" was rated as:

(archive | operational | **IMPORTANT** | critical | mission-critical)

Business Overview

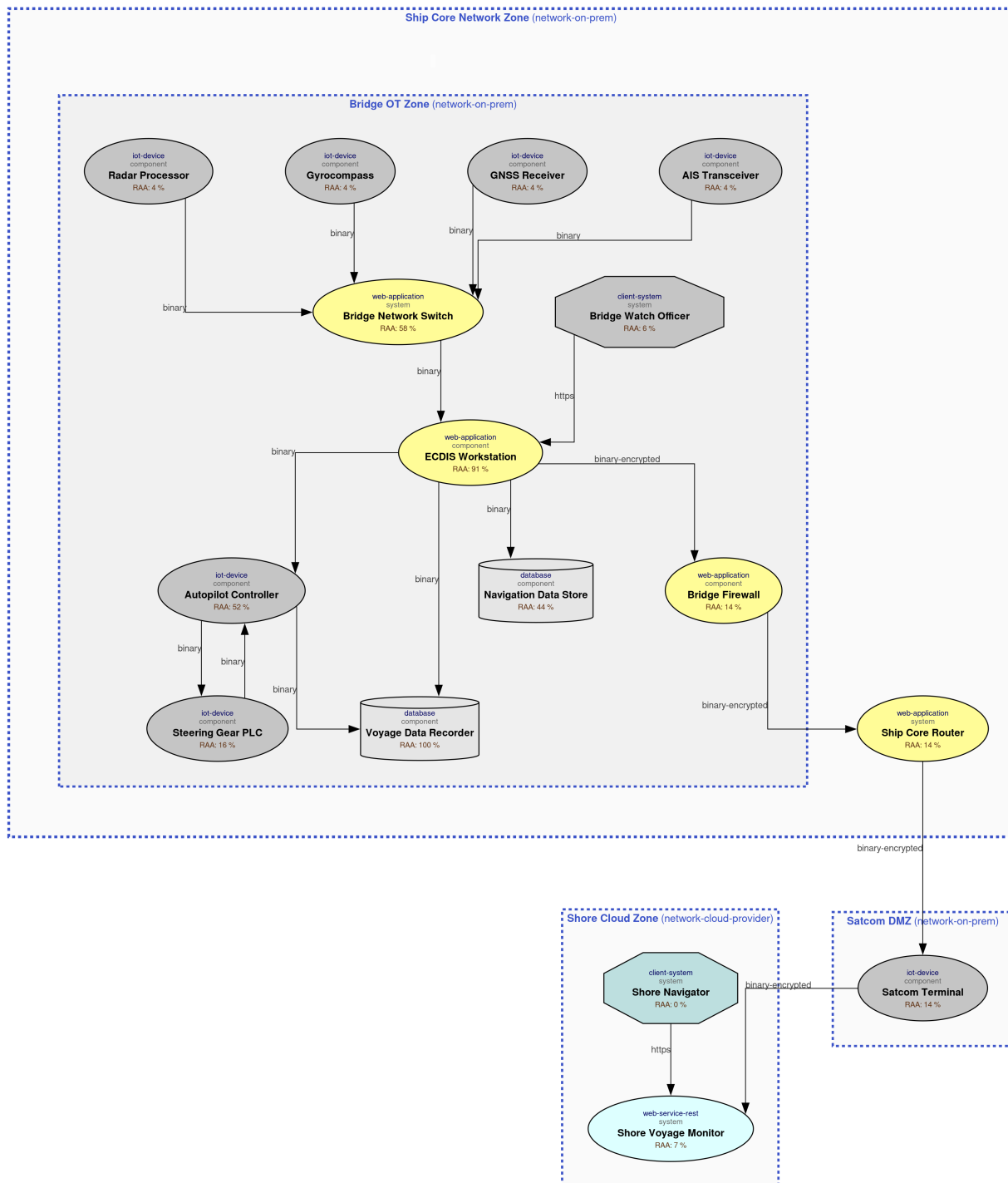
Navigation-at-sea architecture focused on bridge sensors, ECDIS, autopilot control, voyage recording, and the controlled conduit to shore monitoring.

Technical Overview

Bridge systems are physically onboard and segmented from ship IT through a firewall. Navigation sensor feeds are treated as safety-relevant inputs and require authenticity, integrity, and availability review. Shore monitoring is advisory in this scenario and cannot directly command autopilot or steering systems.

Data-Flow Diagram

The following diagram was generated by Threagile based on the model input and gives a high-level overview of the data-flow between technical assets. The RAA value is the calculated *Relative Attacker Attractiveness* in percent. For a full high-resolution version of this diagram please refer to the PNG image file alongside this report.



Security Requirements

This chapter lists the custom security requirements which have been defined for the modeled target.

This list is not complete and regulatory or law relevant security requirements have to be taken into account as well. Also custom individual security requirements might exist for the project.

Abuse Cases

This chapter lists the custom abuse cases which have been defined for the modeled target.

This list is not complete and regulatory or law relevant abuse cases have to be taken into account as well. Also custom individual abuse cases might exist for the project.

Tag Listing

This chapter lists what tags are used by which elements.

data-store

Navigation Data Store, Voyage Data Recorder

human-user

Bridge Watch Officer, Shore Navigator

it-service

Shore Voyage Monitor

network-infrastructure

Bridge Firewall, Bridge Network Switch, Satcom Terminal, Ship Core Router

operational-technology

AIS Transceiver, Autopilot Controller, ECDIS Workstation, GNSS Receiver, Gyrocompass, Radar Processor, Steering Gear PLC, Voyage Data Recorder

STRIDE Classification of Identified Risks

This chapter clusters and classifies the risks by STRIDE categories: In total **17 potential risks** have been identified during the threat modeling process of which **2 in the Spoofing** category, **13 in the Tampering** category, **0 in the Repudiation** category, **1 in the Information Disclosure** category, **0 in the Denial of Service** category, and **1 in the Elevation of Privilege** category.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Spoofing

Medium: **Cross-Site Request Forgery (CSRF)**: 1 / 1 Risk - Exploitation likelihood is *Very Likely* with *Low* impact.

When a web application is accessed via web protocols Cross-Site Request Forgery (CSRF) risks might arise.

Low: **Missing Identity Store**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Tampering

Elevated: **Cross-Site Scripting (XSS)**: 4 / 4 Risks - Exploitation likelihood is *Likely* with *Medium* impact.

For each web application Cross-Site Scripting (XSS) risks might arise. In terms of the overall risk level take other applications running on the same domain into account as well.

Elevated: **SQL/NoSQL-Injection**: 3 / 3 Risks - Exploitation likelihood is *Very Likely* with *Medium* impact.

When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed or stored.

Medium: **Missing Cloud Hardening**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

Cloud components should be hardened according to the cloud vendor best practices. This affects their configuration, auditing, and further areas.

Medium: **Missing Hardening**: 4 / 4 Risks - Exploitation likelihood is *Likely* with *Low* impact.

Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.

Low: **Missing Build Infrastructure**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code

gets developed and built needs to be part of the model.

Repudiation

n/a

Information Disclosure

Low: **Missing Vault (Secret Storage)**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Denial of Service

n/a

Elevation of Privilege

Low: **Missing Network Segmentation**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

Highly sensitive assets and/or datastores residing in the same network segment than other lower sensitive assets (like webserver or content management systems etc.) should be better protected by a network segmentation trust-boundary.

Assignment by Function

This chapter clusters and assigns the risks by functions which are most likely able to check and mitigate them: In total **17 potential risks** have been identified during the threat modeling process of which **0 should be checked by Business Side**, **3 should be checked by Architecture**, **8 should be checked by Development**, and **6 should be checked by Operations**.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Business Side

n/a

Architecture

Low: **Missing Build Infrastructure**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

Include the build infrastructure in the model.

Low: **Missing Identity Store**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

Include an identity store in the model if the application has a login.

Low: **Missing Vault (Secret Storage)**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

Consider using a Vault (Secret Storage) to securely store and access config secrets (like credentials, private keys, client certificates, etc.).

Development

Elevated: **Cross-Site Scripting (XSS)**: 4 / 4 Risks - Exploitation likelihood is *Likely* with *Medium* impact.

Try to encode all values sent back to the browser and also handle DOM-manipulations in a safe way to avoid DOM-based XSS. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Elevated: **SQL/NoSQL-Injection**: 3 / 3 Risks - Exploitation likelihood is *Very Likely* with *Medium* impact.

Try to use parameter binding to be safe from injection vulnerabilities. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Medium: **Cross-Site Request Forgery (CSRF)**: 1 / 1 Risk - Exploitation likelihood is *Very Likely* with *Low* impact.

Try to use anti-CSRF tokens or the double-submit patterns (at least for logged-in requests). When your authentication scheme depends on cookies (like session or token cookies), consider marking them with the same-site flag. When a third-party product is used instead of custom

developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

Operations

Medium: **Missing Cloud Hardening**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Medium* impact.

Apply hardening of all cloud components and services, taking special care to follow the individual risk descriptions (which depend on the cloud provider tags in the model).

Medium: **Missing Hardening**: 4 / 4 Risks - Exploitation likelihood is *Likely* with *Low* impact.

Try to apply all hardening best practices (like CIS benchmarks, OWASP recommendations, vendor recommendations, DevSec Hardening Framework, DBSAT for Oracle databases, and others).

Low: **Missing Network Segmentation**: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

Apply a network segmentation trust-boundary around the highly sensitive assets and/or datastores.

RAA Analysis

For each technical asset the **"Relative Attacker Attractiveness"** (RAA) value was calculated in percent. The higher the RAA, the more interesting it is for an attacker to compromise the asset. The calculation algorithm takes the sensitivity ratings and quantities of stored and processed data into account as well as the communication links of the technical asset. Neighbouring assets to high-value RAA targets might receive an increase in their RAA value when they have a communication link towards that target ("Pivoting-Factor").

The following lists all technical assets sorted by their RAA value from highest (most attacker attractive) to lowest. This list can be used to prioritize on efforts relevant for the most attacker-attractive technical assets:

Technical asset paragraphs are clickable and link to the corresponding chapter.

Voyage Data Recorder: RAA 100%

Imported from UML element 'Voyage Data Recorder' of kind 'database'.

ECDIS Workstation: RAA 91%

Imported from UML element 'ECDIS Workstation' of kind 'component'.

Bridge Network Switch: RAA 58%

Imported from UML element 'Bridge Network Switch' of kind 'node'.

Autopilot Controller: RAA 52%

Imported from UML element 'Autopilot Controller' of kind 'device'.

Navigation Data Store: RAA 44%

Imported from UML element 'Navigation Data Store' of kind 'database'.

Steering Gear PLC: RAA 16%

Imported from UML element 'Steering Gear PLC' of kind 'device'.

Bridge Firewall: RAA 14%

Imported from UML element 'Bridge Firewall' of kind 'component'.

Satcom Terminal: RAA 14%

Imported from UML element 'Satcom Terminal' of kind 'device'.

Ship Core Router: RAA 14%

Imported from UML element 'Ship Core Router' of kind 'node'.

Shore Voyage Monitor: RAA 7%

Imported from UML element 'Shore Voyage Monitor' of kind 'cloud'.

Bridge Watch Officer: RAA 6%

Imported from UML element 'Bridge Watch Officer' of kind 'actor'.

AIS Transceiver: RAA 4%

Imported from UML element 'AIS Transceiver' of kind 'device'.

GNSS Receiver: RAA 4%

Imported from UML element 'GNSS Receiver' of kind 'device'.

Gyrocompass: RAA 4%

Imported from UML element 'Gyrocompass' of kind 'device'.

Radar Processor: RAA 4%

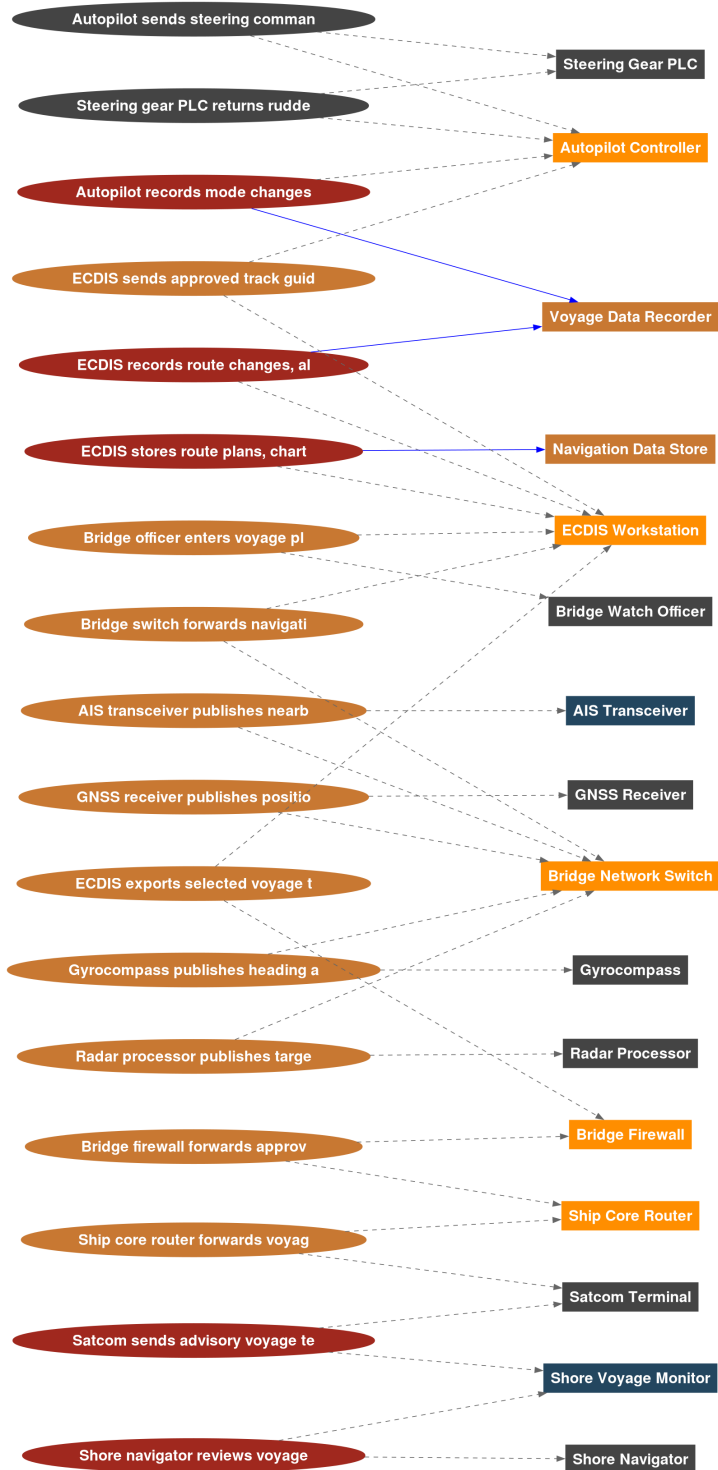
Imported from UML element 'Radar Processor' of kind 'device'.

Shore Navigator: RAA 0%

Imported from UML element 'Shore Navigator' of kind 'actor'.

Data Mapping

The following diagram was generated by Threagile based on the model input and gives a high-level distribution of data assets across technical assets. The color matches the identified data breach probability and risk level (see the "Data Breach Probabilities" chapter for more details). A solid line stands for *data is stored by the asset* and a dashed one means *data is processed by the asset*. For a full high-resolution version of this diagram please refer to the PNG image file alongside this report.



Out-of-Scope Assets: 0 Assets

This chapter lists all technical assets that have been defined as out-of-scope. Each one should be checked in the model whether it should better be included in the overall risk analysis:

Technical asset paragraphs are clickable and link to the corresponding chapter.

No technical assets have been defined as out-of-scope.

Potential Model Failures: 3 / 3 Risks

This chapter lists potential model failures where not all relevant assets have been modeled or the model might itself contain inconsistencies. Each potential model failure should be checked in the model against the architecture design:

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low: Missing Build Infrastructure: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact. The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Low: Missing Identity Store: 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact. The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Low: Missing Vault (Secret Storage): 1 / 1 Risk - Exploitation likelihood is *Unlikely* with *Low* impact.

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Questions: 3 / 3 Questions

This chapter lists custom questions that arose during the threat modeling process.

Flow auth and replay controls?

- answer pending -

Identity and secrets admin?

- answer pending -

Safety critical assets?

- answer pending -

Identified Risks by Vulnerability Category

In total **17 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 0 as high, 7 as elevated, 6 as medium, and 4 as low.**

These risks are distributed across **9 vulnerability categories**. The following sub-chapters of this section describe each identified risk category.

Cross-Site Scripting (XSS): 4 / 4 Risks

Description (Tampering): [CWE 79](#)

For each web application Cross-Site Scripting (XSS) risks might arise. In terms of the overall risk level take other applications running on the same domain into account as well.

Impact

If this risk remains unmitigated, attackers might be able to access individual victim sessions and steal or modify user data.

Detection Logic

In-scope web applications.

Risk Rating

The risk rating depends on the sensitivity of the data processed or stored in the web application.

False Positives

When the technical asset is not accessed via a browser-like component (i.e not by a human user initiating the request that gets passed through all components until it reaches the web application) this can be considered a false positive.

Mitigation (Development): XSS Prevention

Try to encode all values sent back to the browser and also handle DOM-manipulations in a safe way to avoid DOM-based XSS. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V5 - Validation, Sanitization and Encoding Verification Requirements](#)

Cheat Sheet: [Cross Site Scripting Prevention Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Cross-Site Scripting (XSS)** was found **4 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Cross-Site Scripting (XSS) risk at **Bridge Firewall**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@bridge-firewall](#)

Unchecked

Cross-Site Scripting (XSS) risk at **Bridge Network Switch**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@bridge-switch](#)

Unchecked

Cross-Site Scripting (XSS) risk at **ECDIS Workstation**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@ecdis-workstation](#)

Unchecked

Cross-Site Scripting (XSS) risk at **Ship Core Router**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@ship-core-router](#)

Unchecked

SQL/NoSQL-Injection: 3 / 3 Risks

Description (Tampering): [CWE 89](#)

When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed or stored.

Impact

If this risk is unmitigated, attackers might be able to modify SQL/NoSQL queries to steal and modify data and eventually further escalate towards a deeper system penetration via code executions.

Detection Logic

Database accessed via typical database access protocols by in-scope clients.

Risk Rating

The risk rating depends on the sensitivity of the data stored inside the database.

False Positives

Database accesses by queries not consisting of parts controllable by the caller can be considered as false positives after individual review.

Mitigation (Development): SQL/NoSQL-Injection Prevention

Try to use parameter binding to be safe from injection vulnerabilities. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V5 - Validation, Sanitization and Encoding Verification Requirements](#)

Cheat Sheet: [SQL_Injection_Prevention_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **SQL/NoSQL-Injection** was found **3 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

SQL/NoSQL-Injection risk at **Autopilot Controller** against database **Voyage Data Recorder** via **Flow 12**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@autopilot-controller@vdr-recorder@autopilot-controller>flow-12](#)

Unchecked

SQL/NoSQL-Injection risk at **ECDIS Workstation** against database **Navigation Data Store** via **Flow 7**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@ecdis-workstation@navigation-data-store@ecdis-workstation>flow-7](#)

Unchecked

SQL/NoSQL-Injection risk at **ECDIS Workstation** against database **Voyage Data Recorder** via **Flow 11**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@ecdis-workstation@vdr-recorder@ecdis-workstation>flow-11](#)

Unchecked

Cross-Site Request Forgery (CSRF): 1 / 1 Risk

Description (Spoofing): [CWE 352](#)

When a web application is accessed via web protocols Cross-Site Request Forgery (CSRF) risks might arise.

Impact

If this risk remains unmitigated, attackers might be able to trick logged-in victim users into unwanted actions within the web application by visiting an attacker controlled web site.

Detection Logic

In-scope web applications accessed via typical web access protocols.

Risk Rating

The risk rating depends on the integrity rating of the data sent across the communication link.

False Positives

Web applications passing the authentication state via custom headers instead of cookies can eventually be false positives. Also when the web application is not accessed via a browser-like component (i.e not by a human user initiating the request that gets passed through all components until it reaches the web application) this can be considered a false positive.

Mitigation (Development): CSRF Prevention

Try to use anti-CSRF tokens or the double-submit patterns (at least for logged-in requests). When your authentication scheme depends on cookies (like session or token cookies), consider marking them with the same-site flag. When a third-party product is used instead of custom developed software, check if the product applies the proper mitigation and ensure a reasonable patch-level.

ASVS Chapter: [V4 - Access Control Verification Requirements](#)

Cheat Sheet: [Cross-Site_Request_Forgery_Prevention_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Cross-Site Request Forgery (CSRF)** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Cross-Site Request Forgery (CSRF) risk at **ECDIS Workstation** via **Flow 1** from **Bridge Watch Officer**: Exploitation likelihood is *Very Likely* with *Low* impact.

[cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1](#)

Unchecked

Missing Cloud Hardening: 1 / 1 Risk

Description (Tampering): [CWE 1008](#)

Cloud components should be hardened according to the cloud vendor best practices. This affects their configuration, auditing, and further areas.

Impact

If this risk is unmitigated, attackers might access cloud components in an unintended way.

Detection Logic

In-scope cloud components (either residing in cloud trust boundaries or more specifically tagged with cloud provider types).

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

False Positives

Cloud components not running parts of the target architecture can be considered as false positives after individual review.

Mitigation (Operations): Cloud Hardening

Apply hardening of all cloud components and services, taking special care to follow the individual risk descriptions (which depend on the cloud provider tags in the model).

For **Amazon Web Services (AWS)**: Follow the *CIS Benchmark for Amazon Web Services* (see also the automated checks of cloud audit tools like "PacBot", "CloudSploit", "CloudMapper", "ScoutSuite", or "Prowler AWS CIS Benchmark Tool").

For EC2 and other servers running Amazon Linux, follow the *CIS Benchmark for Amazon Linux* and switch to IMDSv2.

For S3 buckets follow the *Security Best Practices for Amazon S3* at

<https://docs.aws.amazon.com/AmazonS3/latest/dev/security-best-practices.html> to avoid accidental leakage.

Also take a look at some of these tools: <https://github.com/toniblyx/my-arsenal-of-aws-security-tools>

For **Microsoft Azure**: Follow the *CIS Benchmark for Microsoft Azure* (see also the automated checks of cloud audit tools like "CloudSploit" or "ScoutSuite").

For **Google Cloud Platform**: Follow the *CIS Benchmark for Google Cloud Computing Platform* (see also the automated checks of cloud audit tools like "*CloudSploit*" or "*ScoutSuite*").

For **Oracle Cloud Platform**: Follow the hardening best practices (see also the automated checks of cloud audit tools like "*CloudSploit*").

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Cloud Hardening** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Cloud Hardening risk at **Shore Cloud Zone**: Exploitation likelihood is *Unlikely* with *Medium* impact.

[missing-cloud-hardening@shore-cloud-zone](#)

Unchecked

Missing Hardening: 4 / 4 Risks

Description (Tampering): [CWE 16](#)

Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.

Impact

If this risk remains unmitigated, attackers might be able to easier attack high-value targets.

Detection Logic

In-scope technical assets with RAA values of 55 % or higher. Generally for high-value targets like datastores, application servers, identity providers and ERP systems this limit is reduced to 40 %

Risk Rating

The risk rating depends on the sensitivity of the data processed or stored in the technical asset.

False Positives

Usually no false positives.

Mitigation (Operations): System Hardening

Try to apply all hardening best practices (like CIS benchmarks, OWASP recommendations, vendor recommendations, DevSec Hardening Framework, DBSAT for Oracle databases, and others).

ASVS Chapter: [V14 - Configuration Verification Requirements](#)

Cheat Sheet: [Attack Surface Analysis Cheat Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Hardening** was found **4 times** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Hardening risk at **Bridge Network Switch**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@bridge-switch](#)

Unchecked

Missing Hardening risk at **ECDIS Workstation**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@ecdis-workstation](#)

Unchecked

Missing Hardening risk at **Navigation Data Store**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@navigation-data-store](#)

Unchecked

Missing Hardening risk at **Voyage Data Recorder**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@vdr-recorder](#)

Unchecked

Missing Build Infrastructure: 1 / 1 Risk

Description (Tampering): [CWE 1127](#)

The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Impact

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model due to critical build infrastructure components missing in the model.

Detection Logic

Models with in-scope custom-developed parts missing in-scope development (code creation) and build infrastructure components (devops-client, sourcecode-repo, build-pipeline, etc.).

Risk Rating

The risk rating depends on the highest sensitivity of the in-scope assets running custom-developed parts.

False Positives

Models not having any custom-developed parts can be considered as false positives after individual review.

Mitigation (Architecture): Build Pipeline Hardening

Include the build infrastructure in the model.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Build Infrastructure** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Build Infrastructure in the threat model (referencing asset **Shore Voyage Monitor** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-build-infrastructure@shore-voyage-monitor](#)

Unchecked

Missing Identity Store: 1 / 1 Risk

Description (Spoofing): [CWE 287](#)

The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Impact

If this risk is unmitigated, attackers might be able to exploit risks unseen in this threat model in the identity provider/store that is currently missing in the model.

Detection Logic

Models with authenticated data-flows authorized via enduser-identity missing an in-scope identity store.

Risk Rating

The risk rating depends on the sensitivity of the enduser-identity authorized technical assets and their data assets processed and stored.

False Positives

Models only offering data/services without any real authentication need can be considered as false positives after individual review.

Mitigation (Architecture): Identity Store

Include an identity store in the model if the application has a login.

ASVS Chapter: [V2 - Authentication Verification Requirements](#)

Cheat Sheet: [Authentication_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Identity Store** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Identity Store in the threat model (referencing asset **Voyage Data Recorder** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-identity-store@vdr-recorder](#)

Unchecked

Missing Network Segmentation: 1 / 1 Risk

Description (Elevation of Privilege): [CWE 1008](#)

Highly sensitive assets and/or datastores residing in the same network segment than other lower sensitive assets (like webserver or content management systems etc.) should be better protected by a network segmentation trust-boundary.

Impact

If this risk is unmitigated, attackers successfully attacking other components of the system might have an easy path towards more valuable targets, as they are not separated by network segmentation.

Detection Logic

In-scope technical assets with high sensitivity and RAA values as well as datastores when surrounded by assets (without a network trust-boundary in-between) which are of type client-system, web-server, web-application, cms, web-service-rest, web-service-soap, build-pipeline, sourcecode-repository, monitoring, or similar and there is no direct connection between these (hence no requirement to be so close to each other).

Risk Rating

Default is low risk. The risk is increased to medium when the asset missing the trust-boundary protection is rated as strictly-confidential or mission-critical.

False Positives

When all assets within the network segmentation trust-boundary are hardened and protected to the same extend as if all were containing/processing highly sensitive data.

Mitigation (Operations): Network Segmentation

Apply a network segmentation trust-boundary around the highly sensitive assets and/or datastores.

ASVS Chapter: [V1 - Architecture, Design and Threat Modeling Requirements](#)

Cheat Sheet: [Attack_Surface_Analysis_Cheat_Sheet](#)

Check

Are recommendations from the linked cheat sheet and referenced ASVS chapter applied?

Risk Findings

The risk **Missing Network Segmentation** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Network Segmentation to further encapsulate and protect **Voyage Data Recorder** against unrelated lower protected assets in the same network segment, which might be easier to compromise by attackers: Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-network-segmentation@vdr-recorder](#)

Unchecked

Missing Vault (Secret Storage): 1 / 1 Risk

Description (Information Disclosure): [CWE 522](#)

In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Impact

If this risk is unmitigated, attackers might be able to easier steal config secrets (like credentials, private keys, client certificates, etc.) once a vulnerability to access files is present and exploited.

Detection Logic

Models without a Vault (Secret Storage).

Risk Rating

The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

False Positives

Models where no technical assets have any kind of sensitive config data to protect can be considered as false positives after individual review.

Mitigation (Architecture): Vault (Secret Storage)

Consider using a Vault (Secret Storage) to securely store and access config secrets (like credentials, private keys, client certificates, etc.).

ASVS Chapter: [V6 - Stored Cryptography Verification Requirements](#)

Cheat Sheet: [Cryptographic_Storage_Cheat_Sheet](#)

Check

Is a Vault (Secret Storage) in place?

Risk Findings

The risk **Missing Vault (Secret Storage)** was found **1 time** in the analyzed architecture to be potentially possible. Each spot should be checked individually by reviewing the implementation whether all controls have been applied properly in order to mitigate each risk.

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Vault (Secret Storage) in the threat model (referencing asset **AIS Transceiver** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-vault@ais-transceiver](#)

Unchecked

Identified Risks by Technical Asset

In total **17 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 0 as high, 7 as elevated, 6 as medium, and 4 as low.**

These risks are distributed across **16 in-scope technical assets**. The following sub-chapters of this section describe each identified risk grouped by technical asset. The RAA value of a technical asset is the calculated "Relative Attacker Attractiveness" value in percent.

Autopilot Controller: 1 / 1 Risk

Description

Imported from UML element 'Autopilot Controller' of kind 'device'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

SQL/NoSQL-Injection risk at **Autopilot Controller** against database **Voyage Data Recorder** via **Flow 12**: Exploitation likelihood is *Very Likely* with *Medium* impact.

[sql-nosql-injection@autopilot-controller@vdr-recorder@autopilot-controller>flow-12](#)

Unchecked

Asset Information

ID:	autopilot-controller
Type:	process
Usage:	business
RAA:	52 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Autopilot records mode changes, Autopilot sends steering comman, ECDIS sends approved track guid, Steering gear PLC returns rudder
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
--------	-------------------

Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 2

Target technical asset names are clickable and link to the corresponding chapter.

Flow 9 (outgoing)

Autopilot sends steering commands to the steering gear PLC.

Target:	Steering Gear PLC
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Autopilot sends steering comman
Data Received:	none

Flow 12 (outgoing)

Autopilot records mode changes, commands, and steering status.

Target:	Voyage Data Recorder
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Autopilot records mode changes

Data Received: none

Incoming Communication Links: 2

Source technical asset names are clickable and link to the corresponding chapter.

Flow 10 (incoming)

Steering gear PLC returns rudder angle and actuator status.

Source:	Steering Gear PLC
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Steering gear PLC returns rudder
Data Sent:	none

Flow 8 (incoming)

ECDIS sends approved track guidance to autopilot.

Source:	ECDIS Workstation
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	ECDIS sends approved track guidance
Data Sent:	none

Bridge Firewall: 1 / 1 Risk

Description

Imported from UML element 'Bridge Firewall' of kind 'component'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Cross-Site Scripting (XSS) risk at **Bridge Firewall**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@bridge-firewall](#)

Unchecked

Asset Information

ID:	bridge-firewall
Type:	process
Usage:	business
RAA:	14 %
Size:	component
Technology:	web-application
Tags:	network-infrastructure
Internet:	false
Machine:	virtual
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	Bridge firewall forwards approv, ECDIS exports selected voyage t
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
Confidentiality:	restricted (rated 3 in scale of 5)

Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 14 (outgoing)

Bridge firewall forwards approved navigation telemetry to ship core.

Target:	Ship Core Router
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Bridge firewall forwards approv
Data Received:	none

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Flow 13 (incoming)

ECDIS exports selected voyage telemetry through the bridge firewall.

Source:	ECDIS Workstation
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false

IP-Filtered: false
Data Received: ECDIS exports selected voyage t
Data Sent: none

Bridge Network Switch: 2 / 2 Risks

Description

Imported from UML element 'Bridge Network Switch' of kind 'node'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Cross-Site Scripting (XSS) risk at **Bridge Network Switch**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@bridge-switch](#)

Unchecked

Medium Risk Severity

Missing Hardening risk at **Bridge Network Switch**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@bridge-switch](#)

Unchecked

Asset Information

ID:	bridge-switch
Type:	process
Usage:	business
RAA:	58 %
Size:	system
Technology:	web-application
Tags:	network-infrastructure
Internet:	false
Machine:	virtual
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	AIS transceiver publishes nearb, Bridge switch forwards navigati, GNSS receiver publishes positio, Gyrocompass publishes heading a, Radar

processor publishes target
Data Stored: none
Formats Accepted: none of the special data formats accepted

Asset Rating

Owner: Imported from UML
Confidentiality: restricted (rated 3 in scale of 5)
Integrity: important (rated 3 in scale of 5)
Availability: important (rated 3 in scale of 5)
CIA-Justification: This asset was inferred automatically from the UML model and needs manual CIA review.

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 6 (outgoing)

Bridge switch forwards navigation sensor data to ECDIS.

Target: ECDIS Workstation
Protocol: binary
Encrypted: false
Authentication: none
Authorization: none
Read-Only: false
Usage: business
Tags: none
VPN: false
IP-Filtered: false
Data Sent: Bridge switch forwards navigati
Data Received: none

Incoming Communication Links: 4

Source technical asset names are clickable and link to the corresponding chapter.

Flow 3 (incoming)

Radar processor publishes target plots and track updates.

Source:	Radar Processor
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Radar processor publishes targe
Data Sent:	none

Flow 5 (incoming)

Gyrocompass publishes heading and rate-of-turn values.

Source:	Gyrocompass
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Gyrocompass publishes heading a
Data Sent:	none

Flow 2 (incoming)

GNSS receiver publishes position, time, and fix quality messages.

Source:	GNSS Receiver
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none

VPN:	false
IP-Filtered:	false
Data Received:	GNSS receiver publishes positio
Data Sent:	none

Flow 4 (incoming)

AIS transceiver publishes nearby vessel and aid-to-navigation messages.

Source:	AIS Transceiver
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	AIS transceiver publishes nearb
Data Sent:	none

ECDIS Workstation: 5 / 5 Risks

Description

Imported from UML element 'ECDIS Workstation' of kind 'component'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

SQL/NoSQL-Injection risk at **ECDIS Workstation** against database **Navigation Data Store** via **Flow 7**: Exploitation likelihood is *Very Likely* with *Medium* impact.

`sql-nosql-injection@ecdis-workstation@navigation-data-store@ecdis-workstation>flow-7`

Unchecked

SQL/NoSQL-Injection risk at **ECDIS Workstation** against database **Voyage Data Recorder** via **Flow 11**: Exploitation likelihood is *Very Likely* with *Medium* impact.

`sql-nosql-injection@ecdis-workstation@vdr-recorder@ecdis-workstation>flow-11`

Unchecked

Cross-Site Scripting (XSS) risk at **ECDIS Workstation**: Exploitation likelihood is *Likely* with *Medium* impact.

`cross-site-scripting@ecdis-workstation`

Unchecked

Medium Risk Severity

Cross-Site Request Forgery (CSRF) risk at **ECDIS Workstation** via **Flow 1** from **Bridge Watch Officer**: Exploitation likelihood is *Very Likely* with *Low* impact.

`cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1`

Unchecked

Missing Hardening risk at **ECDIS Workstation**: Exploitation likelihood is *Likely* with *Low* impact.

`missing-hardening@ecdis-workstation`

Unchecked

Asset Information

ID:	ecdis-workstation
Type:	process
Usage:	business
RAA:	91 %

Size:	component
Technology:	web-application
Tags:	operational-technology
Internet:	false
Machine:	virtual
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	Bridge officer enters voyage pl, Bridge switch forwards navigati, ECDIS exports selected voyage t, ECDIS records route changes, al, ECDIS sends approved track guid, ECDIS stores route plans, chart
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 4

Target technical asset names are clickable and link to the corresponding chapter.

Flow 8 (outgoing)

ECDIS sends approved track guidance to autopilot.

Target:	Autopilot Controller
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business

Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	ECDIS sends approved track guid
Data Received:	none

Flow 7 (outgoing)

ECDIS stores route plans, chart updates, and navigation events.

Target:	Navigation Data Store
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	ECDIS stores route plans, chart
Data Received:	none

Flow 13 (outgoing)

ECDIS exports selected voyage telemetry through the bridge firewall.

Target:	Bridge Firewall
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	ECDIS exports selected voyage t
Data Received:	none

Flow 11 (outgoing)

ECDIS records route changes, alerts, sensor status, and operator actions.

Target:	Voyage Data Recorder
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	enduser-identity-propagation
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	ECDIS records route changes, al
Data Received:	none

Incoming Communication Links: 2

Source technical asset names are clickable and link to the corresponding chapter.

Flow 1 (incoming)

Bridge officer enters voyage plan updates and acknowledges route alarms.

Source:	Bridge Watch Officer
Protocol:	https
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Bridge officer enters voyage pl
Data Sent:	none

Flow 6 (incoming)

Bridge switch forwards navigation sensor data to ECDIS.

Source:	Bridge Network Switch
Protocol:	binary
Encrypted:	false

Authentication: none
Authorization: none
Read-Only: false
Usage: business
Tags: none
VPN: false
IP-Filtered: false
Data Received: Bridge switch forwards navigati
Data Sent: none

Ship Core Router: 1 / 1 Risk

Description

Imported from UML element 'Ship Core Router' of kind 'node'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Elevated Risk Severity

Cross-Site Scripting (XSS) risk at **Ship Core Router**: Exploitation likelihood is *Likely* with *Medium* impact.

[cross-site-scripting@ship-core-router](#)

Unchecked

Asset Information

ID:	ship-core-router
Type:	process
Usage:	business
RAA:	14 %
Size:	system
Technology:	web-application
Tags:	network-infrastructure
Internet:	false
Machine:	virtual
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	Bridge firewall forwards approv, Ship core router forwards voyag
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
Confidentiality:	restricted (rated 3 in scale of 5)

Integrity: important (rated 3 in scale of 5)
Availability: important (rated 3 in scale of 5)
CIA-Justification: This asset was inferred automatically from the UML model and needs manual CIA review.

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 15 (outgoing)

Ship core router forwards voyage telemetry to satcom.

Target:	Satcom Terminal
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Ship core router forwards voyag
Data Received:	none

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Flow 14 (incoming)

Bridge firewall forwards approved navigation telemetry to ship core.

Source:	Bridge Firewall
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false

IP-Filtered: false
Data Received: Bridge firewall forwards approv
Data Sent: none

Navigation Data Store: 1 / 1 Risk

Description

Imported from UML element 'Navigation Data Store' of kind 'database'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Hardening risk at **Navigation Data Store**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@navigation-data-store](#)

Unchecked

Asset Information

ID:	navigation-data-store
Type:	datastore
Usage:	business
RAA:	44 %
Size:	component
Technology:	database
Tags:	data-store
Internet:	false
Machine:	virtual
Encryption:	transparent
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	ECDIS stores route plans, chart
Data Stored:	ECDIS stores route plans, chart
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)

Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Flow 7 (incoming)

ECDIS stores route plans, chart updates, and navigation events.

Source:	ECDIS Workstation
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	ECDIS stores route plans, chart
Data Sent:	none

Voyage Data Recorder: 3 / 3 Risks

Description

Imported from UML element 'Voyage Data Recorder' of kind 'database'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Medium Risk Severity

Missing Hardening risk at **Voyage Data Recorder**: Exploitation likelihood is *Likely* with *Low* impact.

[missing-hardening@vdr-recorder](#)

Unchecked

Low Risk Severity

Missing Identity Store in the threat model (referencing asset **Voyage Data Recorder** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-identity-store@vdr-recorder](#)

Unchecked

Missing Network Segmentation to further encapsulate and protect **Voyage Data Recorder** against unrelated lower protected assets in the same network segment, which might be easier to compromise by attackers: Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-network-segmentation@vdr-recorder](#)

Unchecked

Asset Information

ID:	vdr-recorder
Type:	datastore
Usage:	business
RAA:	100 %
Size:	component
Technology:	database
Tags:	data-store, operational-technology
Internet:	false
Machine:	virtual
Encryption:	transparent
Multi-Tenant:	false

Redundant: false
Custom-Developed: false
Client by Human: false
Data Processed: Autopilot records mode changes, ECDIS records route changes, al
Data Stored: Autopilot records mode changes, ECDIS records route changes, al
Formats Accepted: none of the special data formats accepted

Asset Rating

Owner: Imported from UML
Confidentiality: restricted (rated 3 in scale of 5)
Integrity: important (rated 3 in scale of 5)
Availability: important (rated 3 in scale of 5)
CIA-Justification: This asset was inferred automatically from the UML model and needs manual CIA review.

Incoming Communication Links: 2

Source technical asset names are clickable and link to the corresponding chapter.

Flow 11 (incoming)

ECDIS records route changes, alerts, sensor status, and operator actions.

Source: ECDIS Workstation
Protocol: binary
Encrypted: false
Authentication: none
Authorization: enduser-identity-propagation
Read-Only: false
Usage: business
Tags: none
VPN: false
IP-Filtered: false
Data Received: ECDIS records route changes, al
Data Sent: none

Flow 12 (incoming)

Autopilot records mode changes, commands, and steering status.

Source:	Autopilot Controller
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Autopilot records mode changes
Data Sent:	none

AIS Transceiver: 1 / 1 Risk

Description

Imported from UML element 'AIS Transceiver' of kind 'device'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Vault (Secret Storage) in the threat model (referencing asset **AIS Transceiver** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-vault@ais-transceiver](#)

Unchecked

Asset Information

ID:	ais-transceiver
Type:	process
Usage:	business
RAA:	4 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	AIS transceiver publishes nearb
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
Confidentiality:	restricted (rated 3 in scale of 5)

Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 4 (outgoing)

AIS transceiver publishes nearby vessel and aid-to-navigation messages.

Target:	Bridge Network Switch
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	AIS transceiver publishes nearb
Data Received:	none

Shore Voyage Monitor: 1 / 1 Risk

Description

Imported from UML element 'Shore Voyage Monitor' of kind 'cloud'.

Identified Risks of Asset

Risk finding paragraphs are clickable and link to the corresponding chapter.

Low Risk Severity

Missing Build Infrastructure in the threat model (referencing asset **Shore Voyage Monitor** as an example): Exploitation likelihood is *Unlikely* with *Low* impact.

[missing-build-infrastructure@shore-voyage-monitor](#)

Unchecked

Asset Information

ID:	shore-voyage-monitor
Type:	process
Usage:	business
RAA:	7 %
Size:	system
Technology:	web-service-rest
Tags:	it-service
Internet:	true
Machine:	virtual
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	true
Client by Human:	false
Data Processed:	Satcom sends advisory voyage te, Shore navigator reviews voyage
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
Confidentiality:	restricted (rated 3 in scale of 5)

Integrity: important (rated 3 in scale of 5)
Availability: important (rated 3 in scale of 5)
CIA-Justification: This asset was inferred automatically from the UML model and needs manual CIA review.

Incoming Communication Links: 2

Source technical asset names are clickable and link to the corresponding chapter.

Flow 17 (incoming)

Shore navigator reviews voyage status, route deviations, and safety alerts.

Source:	Shore Navigator
Protocol:	https
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Shore navigator reviews voyage
Data Sent:	none

Flow 16 (incoming)

Satcom sends advisory voyage telemetry to shore monitoring.

Source:	Satcom Terminal
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Satcom sends advisory voyage te
Data Sent:	none

Bridge Watch Officer: 0 / 0 Risks

Description

Imported from UML element 'Bridge Watch Officer' of kind 'actor'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	bridge-watch-officer
Type:	external-entity
Usage:	business
RAA:	6 %
Size:	system
Technology:	client-system
Tags:	human-user
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	true
Data Processed:	Bridge officer enters voyage pl
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 1 (outgoing)

Bridge officer enters voyage plan updates and acknowledges route alarms.

Target:	ECDIS Workstation
Protocol:	https
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Bridge officer enters voyage pl
Data Received:	none

GNSS Receiver: 0 / 0 Risks

Description

Imported from UML element 'GNSS Receiver' of kind 'device'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	gnss-receiver
Type:	process
Usage:	business
RAA:	4 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	GNSS receiver publishes positio
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 2 (outgoing)

GNSS receiver publishes position, time, and fix quality messages.

Target:	Bridge Network Switch
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	GNSS receiver publishes positio
Data Received:	none

Gyrocompass: 0 / 0 Risks

Description

Imported from UML element 'Gyrocompass' of kind 'device'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	gyrocompass
Type:	process
Usage:	business
RAA:	4 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Gyrocompass publishes heading a
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 5 (outgoing)

Gyrocompass publishes heading and rate-of-turn values.

Target:	Bridge Network Switch
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Gyrocompass publishes heading a
Data Received:	none

Radar Processor: 0 / 0 Risks

Description

Imported from UML element 'Radar Processor' of kind 'device'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	radar-processor
Type:	process
Usage:	business
RAA:	4 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Radar processor publishes targe
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 3 (outgoing)

Radar processor publishes target plots and track updates.

Target:	Bridge Network Switch
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Radar processor publishes targe
Data Received:	none

Satcom Terminal: 0 / 0 Risks

Description

Imported from UML element 'Satcom Terminal' of kind 'device'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	satcom-terminal
Type:	process
Usage:	business
RAA:	14 %
Size:	component
Technology:	iot-device
Tags:	network-infrastructure
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Satcom sends advisory voyage te, Ship core router forwards voyag
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML
Confidentiality:	restricted (rated 3 in scale of 5)
Integrity:	important (rated 3 in scale of 5)
Availability:	important (rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 16 (outgoing)

Satcom sends advisory voyage telemetry to shore monitoring.

Target:	Shore Voyage Monitor
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Satcom sends advisory voyage te
Data Received:	none

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Flow 15 (incoming)

Ship core router forwards voyage telemetry to satcom.

Source:	Ship Core Router
Protocol:	binary-encrypted
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Ship core router forwards voyag
Data Sent:	none

Shore Navigator: 0 / 0 Risks

Description

Imported from UML element 'Shore Navigator' of kind 'actor'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	shore-navigator
Type:	external-entity
Usage:	business
RAA:	0 %
Size:	system
Technology:	client-system
Tags:	human-user
Internet:	true
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	true
Data Processed:	Shore navigator reviews voyage
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 17 (outgoing)

Shore navigator reviews voyage status, route deviations, and safety alerts.

Target:	Shore Voyage Monitor
Protocol:	https
Encrypted:	true
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Shore navigator reviews voyage
Data Received:	none

Steering Gear PLC: 0 / 0 Risks

Description

Imported from UML element 'Steering Gear PLC' of kind 'device'.

Identified Risks of Asset

No risks were identified.

Asset Information

ID:	steering-gear-plc
Type:	process
Usage:	business
RAA:	16 %
Size:	component
Technology:	iot-device
Tags:	operational-technology
Internet:	false
Machine:	physical
Encryption:	none
Multi-Tenant:	false
Redundant:	false
Custom-Developed:	false
Client by Human:	false
Data Processed:	Autopilot sends steering comman, Steering gear PLC returns ruddle
Data Stored:	none
Formats Accepted:	none of the special data formats accepted

Asset Rating

Owner:	Imported from UML	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This asset was inferred automatically from the UML model and needs manual CIA review.	

Outgoing Communication Links: 1

Target technical asset names are clickable and link to the corresponding chapter.

Flow 10 (outgoing)

Steering gear PLC returns rudder angle and actuator status.

Target:	Autopilot Controller
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Sent:	Steering gear PLC returns rudder
Data Received:	none

Incoming Communication Links: 1

Source technical asset names are clickable and link to the corresponding chapter.

Flow 9 (incoming)

Autopilot sends steering commands to the steering gear PLC.

Source:	Autopilot Controller
Protocol:	binary
Encrypted:	false
Authentication:	none
Authorization:	none
Read-Only:	false
Usage:	business
Tags:	none
VPN:	false
IP-Filtered:	false
Data Received:	Autopilot sends steering commands
Data Sent:	none

Identified Data Breach Probabilities by Data Asset

In total **17 potential risks** have been identified during the threat modeling process of which **0 are rated as critical, 0 as high, 7 as elevated, 6 as medium, and 4 as low.**

These risks are distributed across **17 data assets**. The following sub-chapters of this section describe the derived data breach probabilities grouped by data asset.

Technical asset names and risk IDs are clickable and link to the corresponding chapter.

Autopilot records mode changes: 4 / 4 Risks

Inferred from the data flow between Autopilot Controller and Voyage Data Recorder.

ID:	flow-data-12		
Usage:	business		
Quantity:	many		
Tags:	none		
Origin:	Autopilot Controller		
Owner:	Voyage Data Recorder		
Confidentiality:	restricted	(rated 3 in scale of 5)	
Integrity:	important	(rated 3 in scale of 5)	
Availability:	important	(rated 3 in scale of 5)	
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.		
Processed by:	Autopilot Controller, Voyage Data Recorder		
Stored by:	Voyage Data Recorder		
Sent via:	Flow 12		
Received via:	none		
Data Breach:	probable		
Data Breach Risks:	This data asset has data breach potential because of 4 remaining risks:		
	Probable: sql-nosql-injection@autopilot-controller@vdr-recorder@autopilot-controller>flow-12		
	Probable: sql-nosql-injection@ecdis-workstation@vdr-recorder@ecdis-workstation>flow-11		
	Improbable: missing-hardening@vdr-recorder		
	Improbable: missing-network-segmentation@vdr-recorder		

ECDIS records route changes, al: 7 / 7 Risks

Inferred from the data flow between ECDIS Workstation and Voyage Data Recorder.

ID:	flow-data-11	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	ECDIS Workstation	
Owner:	Voyage Data Recorder	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	ECDIS Workstation, Voyage Data Recorder	
Stored by:	Voyage Data Recorder	
Sent via:	Flow 11	
Received via:	none	
Data Breach:	probable	
Data Breach Risks:	This data asset has data breach potential because of 7 remaining risks:	

Probable: sql-nosql-injection@autopilot-controller@vdr-recorder@autopilot-controller>flow-12

Probable: sql-nosql-injection@ecdis-workstation@vdr-recorder@ecdis-workstation>flow-11

Possible: cross-site-scripting@ecdis-workstation

Improbable: cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1

Improbable: missing-hardening@ecdis-workstation

Improbable: missing-hardening@vdr-recorder

Improbable: missing-network-segmentation@vdr-recorder

ECDIS stores route plans, chart: 5 / 5 Risks

Inferred from the data flow between ECDIS Workstation and Navigation Data Store.

ID:	flow-data-7	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	ECDIS Workstation	
Owner:	Navigation Data Store	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	ECDIS Workstation, Navigation Data Store	
Stored by:	Navigation Data Store	
Sent via:	Flow 7	
Received via:	none	
Data Breach:	probable	
Data Breach Risks:	This data asset has data breach potential because of 5 remaining risks:	
	Probable: <code>sql-nosql-injection@ecdis-workstation@navigation-data-store@ecdis-workstation>flow-7</code>	
	Possible: <code>cross-site-scripting@ecdis-workstation</code>	
	Improbable: <code>cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1</code>	
	Improbable: <code>missing-hardening@ecdis-workstation</code>	
	Improbable: <code>missing-hardening@navigation-data-store</code>	

Satcom sends advisory voyage te: 1 / 1 Risk

Inferred from the data flow between Satcom Terminal and Shore Voyage Monitor.

ID:	flow-data-16
Usage:	business
Quantity:	many
Tags:	none
Origin:	Satcom Terminal
Owner:	Shore Voyage Monitor
Confidentiality:	restricted (rated 3 in scale of 5)
Integrity:	important (rated 3 in scale of 5)
Availability:	important (rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.
Processed by:	Satcom Terminal, Shore Voyage Monitor
Stored by:	none
Sent via:	Flow 16
Received via:	none
Data Breach:	probable
Data Breach Risks:	This data asset has data breach potential because of 1 remaining risk: Probable: missing-cloud-hardening@shore-cloud-zone

Shore navigator reviews voyage: 1 / 1 Risk

Inferred from the data flow between Shore Navigator and Shore Voyage Monitor.

ID:	flow-data-17		
Usage:	business		
Quantity:	many		
Tags:	none		
Origin:	Shore Navigator		
Owner:	Shore Voyage Monitor		
Confidentiality:	restricted	(rated 3 in scale of 5)	
Integrity:	important	(rated 3 in scale of 5)	
Availability:	important	(rated 3 in scale of 5)	
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.		
Processed by:	Shore Navigator, Shore Voyage Monitor		
Stored by:	none		
Sent via:	Flow 17		
Received via:	none		
Data Breach:	probable		
Data Breach Risks:	This data asset has data breach potential because of 1 remaining risk:		
Probable: missing-cloud-hardening@shore-cloud-zone			

AIS transceiver publishes nearb: 2 / 2 Risks

Inferred from the data flow between AIS Transceiver and Bridge Network Switch.

ID:	flow-data-4	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	AIS Transceiver	
Owner:	Bridge Network Switch	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	AIS Transceiver, Bridge Network Switch	
Stored by:	none	
Sent via:	Flow 4	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 2 remaining risks:	
	Possible: cross-site-scripting@bridge-switch	
	Improbable: missing-hardening@bridge-switch	

Bridge firewall forwards approv: 2 / 2 Risks

Inferred from the data flow between Bridge Firewall and Ship Core Router.

ID:	flow-data-14	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Bridge Firewall	
Owner:	Ship Core Router	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Bridge Firewall, Ship Core Router	
Stored by:	none	
Sent via:	Flow 14	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 2 remaining risks:	
	Possible: cross-site-scripting@bridge-firewall	
	Possible: cross-site-scripting@ship-core-router	

Bridge officer enters voyage pl: 3 / 3 Risks

Inferred from the data flow between Bridge Watch Officer and ECDIS Workstation.

ID:	flow-data-1	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Bridge Watch Officer	
Owner:	ECDIS Workstation	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Bridge Watch Officer, ECDIS Workstation	
Stored by:	none	
Sent via:	Flow 1	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 3 remaining risks:	

Possible: cross-site-scripting@ecdis-workstation

Improbable: cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1

Improbable: missing-hardening@ecdis-workstation

Bridge switch forwards navigati: 5 / 5 Risks

Inferred from the data flow between Bridge Network Switch and ECDIS Workstation.

ID:	flow-data-6	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Bridge Network Switch	
Owner:	ECDIS Workstation	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Bridge Network Switch, ECDIS Workstation	
Stored by:	none	
Sent via:	Flow 6	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 5 remaining risks:	

Possible: cross-site-scripting@bridge-switch

Possible: cross-site-scripting@ecdis-workstation

Improbable: cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1

Improbable: missing-hardening@bridge-switch

Improbable: missing-hardening@ecdis-workstation

ECDIS exports selected voyage t: 4 / 4 Risks

Inferred from the data flow between ECDIS Workstation and Bridge Firewall.

ID:	flow-data-13		
Usage:	business		
Quantity:	many		
Tags:	none		
Origin:	ECDIS Workstation		
Owner:	Bridge Firewall		
Confidentiality:	restricted	(rated 3 in scale of 5)	
Integrity:	important	(rated 3 in scale of 5)	
Availability:	important	(rated 3 in scale of 5)	
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.		
Processed by:	Bridge Firewall, ECDIS Workstation		
Stored by:	none		
Sent via:	Flow 13		
Received via:	none		
Data Breach:	possible		
Data Breach Risks:	This data asset has data breach potential because of 4 remaining risks:		
	Possible: cross-site-scripting@bridge-firewall		
	Possible: cross-site-scripting@ecdis-workstation		
	Improbable: cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1		
	Improbable: missing-hardening@ecdis-workstation		

ECDIS sends approved track guid: 3 / 3 Risks

Inferred from the data flow between ECDIS Workstation and Autopilot Controller.

ID:	flow-data-8	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	ECDIS Workstation	
Owner:	Autopilot Controller	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Autopilot Controller, ECDIS Workstation	
Stored by:	none	
Sent via:	Flow 8	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 3 remaining risks:	

Possible: cross-site-scripting@ecdis-workstation

Improbable: cross-site-request-forgery@ecdis-workstation@bridge-watch-officer>flow-1

Improbable: missing-hardening@ecdis-workstation

GNSS receiver publishes positio: 2 / 2 Risks

Inferred from the data flow between GNSS Receiver and Bridge Network Switch.

ID:	flow-data-2
Usage:	business
Quantity:	many
Tags:	none
Origin:	GNSS Receiver
Owner:	Bridge Network Switch
Confidentiality:	restricted (rated 3 in scale of 5)
Integrity:	important (rated 3 in scale of 5)
Availability:	important (rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.
Processed by:	Bridge Network Switch, GNSS Receiver
Stored by:	none
Sent via:	Flow 2
Received via:	none
Data Breach:	possible
Data Breach Risks:	This data asset has data breach potential because of 2 remaining risks: Possible: cross-site-scripting@bridge-switch Improbable: missing-hardening@bridge-switch

Gyrocompass publishes heading a: 2 / 2 Risks

Inferred from the data flow between Gyrocompass and Bridge Network Switch.

ID:	flow-data-5	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Gyrocompass	
Owner:	Bridge Network Switch	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Bridge Network Switch, Gyrocompass	
Stored by:	none	
Sent via:	Flow 5	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 2 remaining risks:	
	Possible: cross-site-scripting@bridge-switch	
	Improbable: missing-hardening@bridge-switch	

Radar processor publishes target: 2 / 2 Risks

Inferred from the data flow between Radar Processor and Bridge Network Switch.

ID:	flow-data-3	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Radar Processor	
Owner:	Bridge Network Switch	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Bridge Network Switch, Radar Processor	
Stored by:	none	
Sent via:	Flow 3	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 2 remaining risks:	

Possible: [cross-site-scripting@bridge-switch](#)

Improbable: [missing-hardening@bridge-switch](#)

Ship core router forwards voyag: 1 / 1 Risk

Inferred from the data flow between Ship Core Router and Satcom Terminal.

ID:	flow-data-15	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Ship Core Router	
Owner:	Satcom Terminal	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Satcom Terminal, Ship Core Router	
Stored by:	none	
Sent via:	Flow 15	
Received via:	none	
Data Breach:	possible	
Data Breach Risks:	This data asset has data breach potential because of 1 remaining risk:	
	Possible: cross-site-scripting@ship-core-router	

Autopilot sends steering comman: 0 / 0 Risks

Inferred from the data flow between Autopilot Controller and Steering Gear PLC.

ID:	flow-data-9	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Autopilot Controller	
Owner:	Steering Gear PLC	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Autopilot Controller, Steering Gear PLC	
Stored by:	none	
Sent via:	Flow 9	
Received via:	none	
Data Breach:	none	
Data Breach Risks:	This data asset has no data breach potential.	

Steering gear PLC returns rudder: 0 / 0 Risks

Inferred from the data flow between Steering Gear PLC and Autopilot Controller.

ID:	flow-data-10	
Usage:	business	
Quantity:	many	
Tags:	none	
Origin:	Steering Gear PLC	
Owner:	Autopilot Controller	
Confidentiality:	restricted	(rated 3 in scale of 5)
Integrity:	important	(rated 3 in scale of 5)
Availability:	important	(rated 3 in scale of 5)
CIA-Justification:	This data asset was inferred automatically from a UML flow and needs manual refinement.	
Processed by:	Autopilot Controller, Steering Gear PLC	
Stored by:	none	
Sent via:	Flow 10	
Received via:	none	
Data Breach:	none	
Data Breach Risks:	This data asset has no data breach potential.	

Trust Boundaries

In total **5 trust boundaries** have been modeled during the threat modeling process.

Bridge OT Zone

Bridge OT Zone

ID: bridge-ot-zone
Type: [network-on-prem](#)
Tags: none
Assets inside: AIS Transceiver, Autopilot Controller, Bridge Firewall, Bridge Network Switch, Bridge Watch Officer, ECDIS Workstation, GNSS Receiver, Gyrocompass, Navigation Data Store, Radar Processor, Steering Gear PLC, Voyage Data Recorder
Boundaries nested: none

Satcom DMZ

Satcom DMZ

ID: satcom-dmz
Type: [network-on-prem](#)
Tags: none
Assets inside: Satcom Terminal
Boundaries nested: none

Ship Core Network Zone

Ship Core Network Zone

ID: ship-core-network-zone
Type: [network-on-prem](#)
Tags: none
Assets inside: Ship Core Router
Boundaries nested: Bridge OT Zone

Shore Cloud Zone

Shore Cloud Zone

ID: shore-cloud-zone
Type: [network-cloud-provider](#)
Tags: none
Assets inside: Shore Navigator, Shore Voyage Monitor
Boundaries nested: none

Vessel Exterior Zone

Vessel Exterior Zone

ID: vessel-exterior-zone
Type: [network-on-prem](#)
Tags: none
Assets inside: none
Boundaries nested: Bridge OT Zone

Shared Runtimes

In total **0 shared runtime** has been modeled during the threat modeling process.

Risk Rules Checked by Threagile

Threagile Version: 1.0.0

Threagile Build Timestamp: 20240730113903

Threagile Execution Timestamp: 20260529105948

Model Filename: /dev/shm/threagile-input-416753856/threagile-model-3559912473

Model Hash (SHA256): 09d9ac9d4b04f6db1ad8493b4505cb3e91d9579f60c7883d2e55b4d592c8b6e1

Threagile (see <https://threagile.io> for more details) is an open-source toolkit for agile threat modeling, created by Christian Schneider (<https://christian-schneider.net>): It allows to model an architecture with its assets in an agile fashion as a YAML file directly inside the IDE. Upon execution of the Threagile toolkit all standard risk rules (as well as individual custom rules if present) are checked against the architecture model. At the time the Threagile toolkit was executed on the model input file the following risk rules were checked:

Accidental Secret Leak

accidental-secret-leak

STRIDE: Information Disclosure

Description: Sourcecode repositories (including their histories) as well as artifact registries can accidentally contain secrets like checked-in or packaged-in passwords, API tokens, certificates, crypto keys, etc.

Detection: In-scope sourcecode repositories and artifact registries.

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Code Backdooring

code-backdooring

STRIDE: Tampering

Description: For each build-pipeline component Code Backdooring risks might arise where attackers compromise the build-pipeline in order to let backdoored artifacts be shipped into production. Aside from direct code backdooring this includes backdooring of dependencies and even of more lower-level build infrastructure, like backdooring compilers (similar to what the XcodeGhost malware did) or dependencies.

Detection: In-scope development relevant technical assets which are either accessed by out-of-scope unmanaged developer clients and/or are directly accessed by any kind of internet-located (non-VPN) component or are themselves directly located on the internet.

Rating: The risk rating depends on the confidentiality and integrity rating of the code being handled and deployed as well as the placement/calling of this technical asset on/from the internet.

Container Base Image Backdooring

container-baseimage-backdooring

STRIDE: Tampering

Description: When a technical asset is built using container technologies, Base Image Backdooring risks might arise where base images and other layers used contain vulnerable components or backdoors.

Detection: In-scope technical assets running as containers.

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets.

Container Platform Escape

container-platform-escape

STRIDE: Elevation of Privilege

Description: Container platforms are especially interesting targets for attackers as they host big parts of a containerized runtime infrastructure. When not configured and operated with security best practices in mind, attackers might exploit a vulnerability inside an container and escape towards the platform as highly privileged users. These scenarios might give attackers capabilities to attack every other container as owning the container platform (via container escape attacks) equals to owning every container.

Detection: In-scope container platforms.

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Cross-Site Request Forgery (CSRF)

cross-site-request-forgery

STRIDE: Spoofing

Description: When a web application is accessed via web protocols Cross-Site Request Forgery (CSRF) risks might arise.

Detection: In-scope web applications accessed via typical web access protocols.

Rating: The risk rating depends on the integrity rating of the data sent across the communication link.

Cross-Site Scripting (XSS)

cross-site-scripting

STRIDE: Tampering

Description: For each web application Cross-Site Scripting (XSS) risks might arise. In terms of the overall risk level take other applications running on the same domain into account as well.

Detection: In-scope web applications.

Rating: The risk rating depends on the sensitivity of the data processed or stored in the web application.

DoS-risky Access Across Trust-Boundary

dos-risky-access-across-trust-boundary

STRIDE: Denial of Service

Description: Assets accessed across trust boundaries with critical or mission-critical availability rating are more prone to Denial-of-Service (DoS) risks.

Detection: In-scope technical assets (excluding load-balancer) with availability rating of critical or higher which have incoming data-flows across a network trust-boundary (excluding devops usage).

Rating: Matching technical assets with availability rating of critical or higher are at low risk. When the availability rating is mission-critical and neither a VPN nor IP filter for the incoming data-flow nor redundancy for the asset is applied, the risk-rating is considered medium.

Incomplete Model**incomplete-model**

STRIDE: Information Disclosure

Description: When the threat model contains unknown technologies or transfers data over unknown protocols, this is an indicator for an incomplete model.

Detection: All technical assets and communication links with technology type or protocol type specified as unknown.

Rating: low

LDAP-Injection**ldap-injection**

STRIDE: Tampering

Description: When an LDAP server is accessed LDAP-Injection risks might arise. The risk rating depends on the sensitivity of the LDAP server itself and of the data assets processed or stored.

Detection: In-scope clients accessing LDAP servers via typical LDAP access protocols.

Rating: The risk rating depends on the sensitivity of the LDAP server itself and of the data assets processed or stored.

Missing Authentication**missing-authentication**

STRIDE: Elevation of Privilege

Description: Technical assets (especially multi-tenant systems) should authenticate incoming requests when the asset processes or stores sensitive data.

Detection: In-scope technical assets (except load-balancer, reverse-proxy, service-registry, waf, ids, and ips and in-process calls) should authenticate incoming requests when the asset processes or stores sensitive data. This is especially the case for all multi-tenant assets (there even non-sensitive ones).

Rating: The risk rating (medium or high) depends on the sensitivity of the data sent across

the communication link. Monitoring callers are exempted from this risk.

Missing Two-Factor Authentication (2FA)

missing-authentication-second-factor

STRIDE: Elevation of Privilege

Description: Technical assets (especially multi-tenant systems) should authenticate incoming requests with two-factor (2FA) authentication when the asset processes or stores highly sensitive data (in terms of confidentiality, integrity, and availability) and is accessed by humans.

Detection: In-scope technical assets (except load-balancer, reverse-proxy, waf, ids, and ips) should authenticate incoming requests via two-factor authentication (2FA) when the asset processes or stores highly sensitive data (in terms of confidentiality, integrity, and availability) and is accessed by a client used by a human user.

Rating: medium

Missing Build Infrastructure

missing-build-infrastructure

STRIDE: Tampering

Description: The modeled architecture does not contain a build infrastructure (devops-client, sourcecode-repo, build-pipeline, etc.), which might be the risk of a model missing critical assets (and thus not seeing their risks). If the architecture contains custom-developed parts, the pipeline where code gets developed and built needs to be part of the model.

Detection: Models with in-scope custom-developed parts missing in-scope development (code creation) and build infrastructure components (devops-client, sourcecode-repo, build-pipeline, etc.).

Rating: The risk rating depends on the highest sensitivity of the in-scope assets running custom-developed parts.

Missing Cloud Hardening

missing-cloud-hardening

STRIDE: Tampering

Description: Cloud components should be hardened according to the cloud vendor best practices. This affects their configuration, auditing, and further areas.

Detection: In-scope cloud components (either residing in cloud trust boundaries or more specifically tagged with cloud provider types).

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Missing File Validation

missing-file-validation

STRIDE: Spoofing

- Description:** When a technical asset accepts files, these input files should be strictly validated about filename and type.
- Detection:** In-scope technical assets with custom-developed code accepting file data formats.
- Rating:** The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Missing Hardening

missing-hardening

- STRIDE:** Tampering
- Description:** Technical assets with a Relative Attacker Attractiveness (RAA) value of 55 % or higher should be explicitly hardened taking best practices and vendor hardening guides into account.
- Detection:** In-scope technical assets with RAA values of 55 % or higher. Generally for high-value targets like datastores, application servers, identity providers and ERP systems this limit is reduced to 40 %
- Rating:** The risk rating depends on the sensitivity of the data processed or stored in the technical asset.

Missing Identity Propagation

missing-identity-propagation

- STRIDE:** Elevation of Privilege
- Description:** Technical assets (especially multi-tenant systems), which usually process data for endusers should authorize every request based on the identity of the enduser when the data flow is authenticated (i.e. non-public). For DevOps usages at least a technical-user authorization is required.
- Detection:** In-scope service-like technical assets which usually process data based on enduser requests, if authenticated (i.e. non-public), should authorize incoming requests based on the propagated enduser identity when their rating is sensitive. This is especially the case for all multi-tenant assets (there even less-sensitive rated ones). DevOps usages are exempted from this risk.
- Rating:** The risk rating (medium or high) depends on the confidentiality, integrity, and availability rating of the technical asset.

Missing Identity Provider Isolation

missing-identity-provider-isolation

- STRIDE:** Elevation of Privilege
- Description:** Highly sensitive identity provider assets and their identity datastores should be isolated from other assets by their own network segmentation trust-boundary (execution-environment boundaries do not count as network isolation).
- Detection:** In-scope identity provider assets and their identity datastores when surrounded by other (not identity-related) assets (without a network trust-boundary in-between).

This risk is especially prevalent when other non-identity related assets are within the same execution environment (i.e. same database or same application server).

Rating: Default is high impact. The impact is increased to very-high when the asset missing the trust-boundary protection is rated as strictly-confidential or mission-critical.

Missing Identity Store

missing-identity-store

STRIDE: Spoofing

Description: The modeled architecture does not contain an identity store, which might be the risk of a model missing critical assets (and thus not seeing their risks).

Detection: Models with authenticated data-flows authorized via enduser-identity missing an in-scope identity store.

Rating: The risk rating depends on the sensitivity of the enduser-identity authorized technical assets and their data assets processed and stored.

Missing Network Segmentation

missing-network-segmentation

STRIDE: Elevation of Privilege

Description: Highly sensitive assets and/or datastores residing in the same network segment than other lower sensitive assets (like webserver or content management systems etc.) should be better protected by a network segmentation trust-boundary.

Detection: In-scope technical assets with high sensitivity and RAA values as well as datastores when surrounded by assets (without a network trust-boundary in-between) which are of type client-system, web-server, web-application, cms, web-service-rest, web-service-soap, build-pipeline, sourcecode-repository, monitoring, or similar and there is no direct connection between these (hence no requirement to be so close to each other).

Rating: Default is low risk. The risk is increased to medium when the asset missing the trust-boundary protection is rated as strictly-confidential or mission-critical.

Missing Vault (Secret Storage)

missing-vault

STRIDE: Information Disclosure

Description: In order to avoid the risk of secret leakage via config files (when attacked through vulnerabilities being able to read files like Path-Traversal and others), it is best practice to use a separate hardened process with proper authentication, authorization, and audit logging to access config secrets (like credentials, private keys, client certificates, etc.). This component is usually some kind of Vault.

Detection: Models without a Vault (Secret Storage).

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Missing Vault Isolation

missing-vault-isolation

STRIDE: Elevation of Privilege

Description: Highly sensitive vault assets and their datastores should be isolated from other assets by their own network segmentation trust-boundary (execution-environment boundaries do not count as network isolation).

Detection: In-scope vault assets when surrounded by other (not vault-related) assets (without a network trust-boundary in-between). This risk is especially prevalent when other non-vault related assets are within the same execution environment (i.e. same database or same application server).

Rating: Default is medium impact. The impact is increased to high when the asset missing the trust-boundary protection is rated as strictly-confidential or mission-critical.

Missing Web Application Firewall (WAF)

missing-waf

STRIDE: Tampering

Description: To have a first line of filtering defense, security architectures with web-services or web-applications should include a WAF in front of them. Even though a WAF is not a replacement for security (all components must be secure even without a WAF) it adds another layer of defense to the overall system by delaying some attacks and having easier attack alerting through it.

Detection: In-scope web-services and/or web-applications accessed across a network trust boundary not having a Web Application Firewall (WAF) in front of them.

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Mixed Targets on Shared Runtime

mixed-targets-on-shared-runtime

STRIDE: Elevation of Privilege

Description: Different attacker targets (like frontend and backend/datastore components) should not be running on the same shared (underlying) runtime.

Detection: Shared runtime running technical assets of different trust-boundaries is at risk. Also mixing backend/datastore with frontend components on the same shared runtime is considered a risk.

Rating: The risk rating (low or medium) depends on the confidentiality, integrity, and availability rating of the technical asset running on the shared runtime.

Path-Traversal

path-traversal

STRIDE: Information Disclosure

Description: When a filesystem is accessed Path-Traversal or Local-File-Inclusion (LFI) risks might arise. The risk rating depends on the sensitivity of the technical asset itself

and of the data assets processed or stored.

Detection: Filesystems accessed by in-scope callers.

Rating: The risk rating depends on the sensitivity of the data stored inside the technical asset.

Push instead of Pull Deployment

push-instead-of-pull-deployment

STRIDE: Tampering

Description: When comparing push-based vs. pull-based deployments from a security perspective, pull-based deployments improve the overall security of the deployment targets. Every exposed interface of a production system to accept a deployment increases the attack surface of the production system, thus a pull-based approach exposes less attack surface relevant interfaces.

Detection: Models with build pipeline components accessing in-scope targets of deployment (in a non-readonly way) which are not build-related components themselves.

Rating: The risk rating depends on the highest sensitivity of the deployment targets running custom-developed parts.

Search-Query Injection

search-query-injection

STRIDE: Tampering

Description: When a search engine server is accessed Search-Query Injection risks might arise.

Detection: In-scope clients accessing search engine servers via typical search access protocols.

Rating: The risk rating depends on the sensitivity of the search engine server itself and of the data assets processed or stored.

Server-Side Request Forgery (SSRF)

server-side-request-forgery

STRIDE: Information Disclosure

Description: When a server system (i.e. not a client) is accessing other server systems via typical web protocols Server-Side Request Forgery (SSRF) or Local-File-Inclusion (LFI) or Remote-File-Inclusion (RFI) risks might arise.

Detection: In-scope non-client systems accessing (using outgoing communication links) targets with either HTTP or HTTPS protocol.

Rating: The risk rating (low or medium) depends on the sensitivity of the data assets receivable via web protocols from targets within the same network trust-boundary as well on the sensitivity of the data assets receivable via web protocols from the target asset itself. Also for cloud-based environments the exploitation impact is at least medium, as cloud backend services can be attacked via SSRF.

Service Registry Poisoning

service-registry-poisoning**STRIDE:** Spoofing**Description:** When a service registry used for discovery of trusted service endpoints Service Registry Poisoning risks might arise.**Detection:** In-scope service registries.**Rating:** The risk rating depends on the sensitivity of the technical assets accessing the service registry as well as the data assets processed or stored.**SQL/NoSQL-Injection****sql-nosql-injection****STRIDE:** Tampering**Description:** When a database is accessed via database access protocols SQL/NoSQL-Injection risks might arise. The risk rating depends on the sensitivity technical asset itself and of the data assets processed or stored.**Detection:** Database accessed via typical database access protocols by in-scope clients.**Rating:** The risk rating depends on the sensitivity of the data stored inside the database.**Unchecked Deployment****unchecked-deployment****STRIDE:** Tampering**Description:** For each build-pipeline component Unchecked Deployment risks might arise when the build-pipeline does not include established DevSecOps best-practices. DevSecOps best-practices scan as part of CI/CD pipelines for vulnerabilities in source- or byte-code, dependencies, container layers, and dynamically against running test systems. There are several open-source and commercial tools existing in the categories DAST, SAST, and IAST.**Detection:** All development-relevant technical assets.**Rating:** The risk rating depends on the highest rating of the technical assets and data assets processed by deployment-receiving targets.**Unencrypted Technical Assets****unencrypted-asset****STRIDE:** Information Disclosure**Description:** Due to the confidentiality rating of the technical asset itself and/or the processed data assets this technical asset must be encrypted. The risk rating depends on the sensitivity technical asset itself and of the data assets stored.**Detection:** In-scope unencrypted technical assets (excluding reverse-proxy, load-balancer, waf, ids, ips and embedded components like library) storing data assets rated at least as confidential or critical. For technical assets storing data assets rated as strictly-confidential or mission-critical the encryption must be of type data-with-enduser-individual-key.

Rating: Depending on the confidentiality rating of the stored data-assets either medium or high risk.

Unencrypted Communication

unencrypted-communication

STRIDE: Information Disclosure

Description: Due to the confidentiality and/or integrity rating of the data assets transferred over the communication link this connection must be encrypted.

Detection: Unencrypted technical communication links of in-scope technical assets (excluding monitoring traffic as well as local-file-access and in-process-library-call) transferring sensitive data.

Rating: Depending on the confidentiality rating of the transferred data-assets either medium or high risk.

Unguarded Access From Internet

unguarded-access-from-internet

STRIDE: Elevation of Privilege

Description: Internet-exposed assets must be guarded by a protecting service, application, or reverse-proxy.

Detection: In-scope technical assets (excluding load-balancer) with confidentiality rating of confidential (or higher) or with integrity rating of critical (or higher) when accessed directly from the internet. All web-server, web-application, reverse-proxy, waf, and gateway assets are exempted from this risk when they do not consist of custom developed code and the data-flow only consists of HTTP or FTP protocols. Access from monitoring systems as well as VPN-protected connections are exempted.

Rating: The matching technical assets are at low risk. When either the confidentiality rating is strictly-confidential or the integrity rating is mission-critical, the risk-rating is considered medium. For assets with RAA values higher than 40 % the risk-rating increases.

Unguarded Direct Datastore Access

unguarded-direct-datastore-access

STRIDE: Elevation of Privilege

Description: Datastores accessed across trust boundaries must be guarded by some protecting service or application.

Detection: In-scope technical assets of type datastore (except identity-store-ldap when accessed from identity-provider and file-server when accessed via file transfer protocols) with confidentiality rating of confidential (or higher) or with integrity rating of critical (or higher) which have incoming data-flows from assets outside across a network trust-boundary. DevOps config and deployment access is excluded from this risk.

Rating: The matching technical assets are at low risk. When either the confidentiality rating is strictly-confidential or the integrity rating is mission-critical, the risk-rating is considered medium. For assets with RAA values higher than 40 % the risk-rating increases.

Unnecessary Communication Link

unnecessary-communication-link

STRIDE: Elevation of Privilege

Description: When a technical communication link does not send or receive any data assets, this is an indicator for an unnecessary communication link (or for an incomplete model).

Detection: In-scope technical assets' technical communication links not sending or receiving any data assets.

Rating: low

Unnecessary Data Asset

unnecessary-data-asset

STRIDE: Elevation of Privilege

Description: When a data asset is not processed or stored by any data assets and also not transferred by any communication links, this is an indicator for an unnecessary data asset (or for an incomplete model).

Detection: Modelled data assets not processed or stored by any data assets and also not transferred by any communication links.

Rating: low

Unnecessary Data Transfer

unnecessary-data-transfer

STRIDE: Elevation of Privilege

Description: When a technical asset sends or receives data assets, which it neither processes or stores this is an indicator for unnecessarily transferred data (or for an incomplete model). When the unnecessarily transferred data assets are sensitive, this poses an unnecessary risk of an increased attack surface.

Detection: In-scope technical assets sending or receiving sensitive data assets which are neither processed nor stored by the technical asset are flagged with this risk. The risk rating (low or medium) depends on the confidentiality, integrity, and availability rating of the technical asset. Monitoring data is exempted from this risk.

Rating: The risk assessment is depending on the confidentiality and integrity rating of the transferred data asset either low or medium.

Unnecessary Technical Asset

unnecessary-technical-asset

STRIDE: Elevation of Privilege

Description: When a technical asset does not process or store any data assets, this is an

indicator for an unnecessary technical asset (or for an incomplete model). This is also the case if the asset has no communication links (either outgoing or incoming).

Detection: Technical assets not processing or storing any data assets.

Rating: low

Untrusted Deserialization

untrusted-deserialization

STRIDE: Tampering

Description: When a technical asset accepts data in a specific serialized form (like Java or .NET serialization), Untrusted Deserialization risks might arise.

Detection: In-scope technical assets accepting serialization data formats (including EJB and RMI protocols).

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data assets processed and stored.

Wrong Communication Link Content

wrong-communication-link-content

STRIDE: Information Disclosure

Description: When a communication link is defined as readonly, but does not receive any data asset, or when it is defined as not readonly, but does not send any data asset, it is likely to be a model failure.

Detection: Communication links with inconsistent data assets being sent/received not matching their readonly flag or otherwise inconsistent protocols not matching the target technology type.

Rating: low

Wrong Trust Boundary Content

wrong-trust-boundary-content

STRIDE: Elevation of Privilege

Description: When a trust boundary of type network-policy-namespace-isolation contains non-container assets it is likely to be a model failure.

Detection: Trust boundaries which should only contain containers, but have different assets inside.

Rating: low

XML External Entity (XXE)

xml-external-entity

STRIDE: Information Disclosure

Description: When a technical asset accepts data in XML format, XML External Entity (XXE) risks might arise.

Detection: In-scope technical assets accepting XML data formats.

Rating: The risk rating depends on the sensitivity of the technical asset itself and of the data

assets processed and stored. Also for cloud-based environments the exploitation impact is at least medium, as cloud backend services can be attacked via SSRF (and XXE vulnerabilities are often also SSRF vulnerabilities).

Disclaimer

threatmod-automation conducted this threat analysis using the open-source Threagile toolkit on the applications and systems that were modeled as of this report's date. Information security threats are continually changing, with new vulnerabilities discovered on a daily basis, and no application can ever be 100% secure no matter how much threat modeling is conducted. It is recommended to execute threat modeling and also penetration testing on a regular basis (for example yearly) to ensure a high ongoing level of security and constantly check for new attack vectors.

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In this particular project, a timebox approach was used to define the analysis effort. This means that the author allotted a prearranged amount of time to identify and document threats. Because of this, there is no guarantee that all possible threats and risks are discovered. Furthermore, the analysis applies to a snapshot of the current state of the modeled architecture (based on the architecture information provided by the customer) at the examination time.

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